

Expand and Factorise - Review

Learning Objective	Apply algebraic properties to rearrange, expand and factorise linear expressions
Today We Are	Expanding and Factorising
Indicators of Progress	• I can expand brackets • I can factorise

Expand

Multiply the outside term (number or letter) by every term inside the bracket:

Expand:

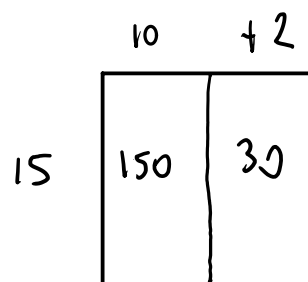
$$4(x + 5)$$

Multiply 4 by every term inside the bracket:

$$= 4x + 20$$

Example 1 (Numbers only)

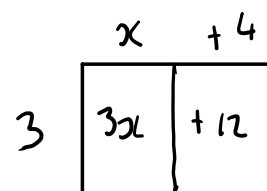
$$\begin{aligned} 15 \times 12 & \quad 15(10 + 2) \\ & = 150 + 30 \\ & = 180 \end{aligned}$$



Example 2 (Algebra)

a) Expand $3(x + 4)$

$$= 3x + 12$$



$$= 2x + 12$$

b) Expand $2(5 + x)$

$$= 10 + 2x$$

	5	+ x
2	10	2x

c) Expand $-2x(x + 8)$

$$= -2x^2 - 16x$$

	x	+ 8
-2x	-2x^2	-16x

d) Expand and simplify $2(x - 9) + 4x$

$$= 2x - 18 + 4x$$

$$= 6x - 18$$

e) Expand and simplify $5(x + 2) + 8(2x + 3)$

$$= 5x + 10 + 16x + 24$$

$$= 21x + 34$$

Factorise

The opposite of expanding.

Factorise:

$$8x + 12$$

Highest Common Factor = 4

$$= 4(2x + 3)$$

Example 1

Factorise $3x + 12$

$$3(x + 4)$$

$$\begin{array}{c} x \quad + 4 \\ 3 \left[\begin{array}{c|c} 3x & +12 \end{array} \right] \\ 1, 3 \quad 1, 2, 3, 4, 6, 12 \end{array}$$

Example 2

Factorise $10x + 15$ HCF = 5

$$5(2x + 3)$$

Example 3

Factorise $16xyz - 4xy + 8xz$

$$4x(4yz - y + 2z)$$

$$16 \rightarrow 1, 2, 4, 8, 16$$

$$4 \rightarrow 1, 2, 4$$

$$8 \rightarrow 1, 2, 4, 8$$



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